

From Code to Careers: Linking Open-Source Software Signals to AI Skill Demand in the Labour Market

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Abstract

The abstract stands alone as a very short version of the dissertation.

The abstract should state the scope and principal objectives of the project, describe the methods, summarize the results and state the principal conclusions.
(Max. 500 words.)

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Acknowledgment

To whoever has the patience to read this :-)

This section is customary, but not obligatory. It makes a brief statement of thanks to those who have helped.

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Chapter 1

Introduction

[1] In recent times, we are seeing that skills and job requirements are evolving rapidly. As the culmination of various effects, including the development of new technologies and evolving consumer preferences, the labour force must adapt in order to satisfy the new requirements. Given the historic pace of labour demand shifts, traditional methods of forecasting these changes are not well-equipped to handle this blistering, and accelerating, rate of change.

By these so-called "Traditional" methods, I mean that which accompany government or firm-level analysis of the skills market. These depend upon data which are limited in the following ways:

1. They use structured data
2. They collect the data periodically (e.g. Annually / Quarterly)

An important feature of these methods is their approach to modelling the dynamics influencing labour demand. Of key importance is the impact of innovation, and how new technology is estimated to affect skill requirements.

This paper aims to investigate whether Open Source Software (OSS) innovation Signals, derived from unstructured text data, can meaningfully contribute to the forecast of AI-related skill requirements present in Job Postings.

To break this down, there are a few key ideas being explored:

1. Can open source software data represent information about Innovation in the AI sector?
2. Can AI-related skill requirements be modelled the textual data within Job Postings?
3. Can we integrate the two data channels to produce a quantified measure of their relationships?

Put simply, if we can measure how similar an OSS repository (innovation-source) is to Job Postings, in terms of the skill-profile and concepts involved, then we can represent how this innovation-source has diffused over time (i.e. by the date of Job Posting). By relating this to the temporal development of the innovation-source, as represented by some other measure of the OSS repository’s history, we can the development-diffusion timeline of that particular innovation.

If true, this insight has tangible benefits for the practice forecasting skill demand in the AI sector, within which open-source software is a prolific source of innovation. Moreover, Job Postings & OSS data are high-frequency and accessible, meaning that more granular and fine-tuned predictions can be produced in effective real-time.

1.1 Aims & Objectives

1.1.1 Primary Aim

To develop and evaluate an AI-related skill demand forecasting methodology, integrating innovation signals (from Open-Source Software in particular) and using job ads as a proxy for skill demand.

1.1.2 Objectives

1. **Data Collection:** Collect & collate data for both channels in:
 - (a) Open-source Software (GitHub): *textual content & historic measure of development*
 - (b) Job Postings (e.g. LinkedIn, Indeed) - *content & date-posted*
2. **Representation:** Develop a high-quality representation method for integrating textual data from both channels, involving:
 - (a) **Build a Baseline:** A weak, keyword-based mechanism.
 - (b) **Advanced Representation:** An intelligent and robust method for representing text as embeddings.
 - (c) **Experimentation:** Fine tune and alter the components of embedding model, evaluating against the baseline.
3. **Time series:** Investigate Relationship of derived time series, looking at Granger causality values and Pearson correlations.
4. **Discuss:** Critically evaluate the derived results, With reflections on the benefits and limitations of the approach.

1.2 Roadmap

In the following chapter, I discuss related pieces of work and how their findings informed the paradigm for this project's implementation. Given the domain intersectional nature of this paper, works that are "related" include those on the topics of economic theory, natural language processing and AI itself.

Following this, the implementation of this project is detailed. Firstly, this concerns how data was acquired and preprocessed. Thereafter, the primary component of the pipeline is discussed - namely, the representation pipeline. Within this, I demonstrate both how the baseline model and the representation model are built, discussing the involved components and how experimentation can be conducted to find optimal constructions. The subsequent section demonstrates the downstream task of these representation results - in the context of time series analysis, the findings of our representation pipeline are presented alongside the data of GitHub Stars. Having demonstrated the function of the entire pipeline, this next section therefore details the results and rationale of several sets of experiments.

What thereafter follows is a critical evaluation of the process, discussing the project's limitations and its successes, as well as the opportunities for future work. Finally, We discuss in the conclusion the outcomes and results of this paper, and their implications.

Chapter 2

Literature review

2.1 Economic Background

2.1.1 Labour (Employment) Demand & Dynamics

Labour demand is defined as the number of people that employers are willing to employ at a given wage during a specific time period [1].

As a result of various economic phenomena, Labour demand is dynamic and frequently changes: Firm-level performance, competition and policies in minimum-wage or taxation are among the many factors that can influence a firm's capacity to employ. At the microeconomic (firm/individual) scale, much of the negative effects of labour demand equilibria changes Are expected in a free-market economy.

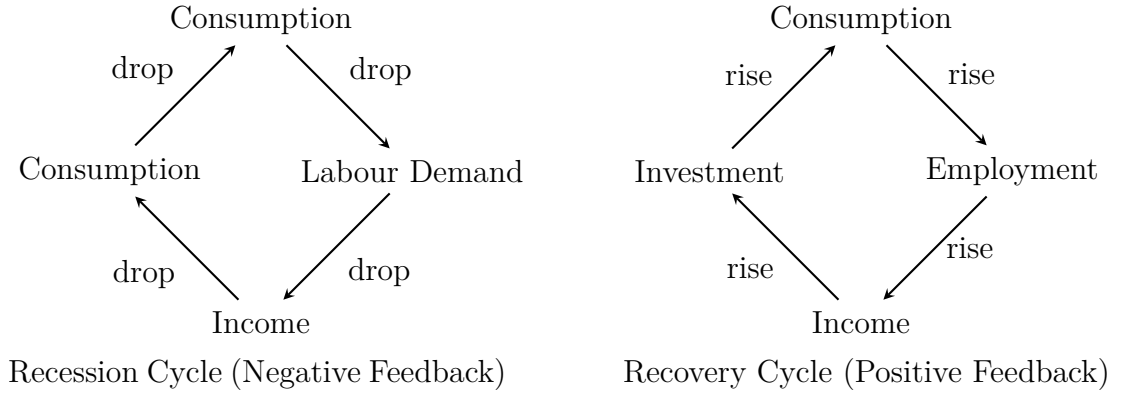
There are, however, systematic shocks that would affect the labour demand on a macroeconomic (economy-wide) sense. These can be deeply damaging to the proper functioning of that economy. Examples of such systematic shocks include war, pandemics or financial crises. Broadly, these shocks will have effects that extend to beyond labour demand shifts, since the economy is composed of an unfathomably complex network of cause-effect relationships.

Such a consequence of this interconnectedness is the existence of feedback loops within the economy, as the standard Keynesian representation of recessions depicts [2] (see Figure 2.1):

2.1.2 Theories of Innovation

Another driver of shifts in Labour demand is technology, or more generally, as innovation. Classical economic theory and some research suggests that technological development & adoption can be seen as a labour augmentation device, contributing positively to long term growth and employment [2], [3]. An opposing view, most famously represented by the example of the Luddites [4], depicts

Figure 2.1: The inter-connected Economy



technological advancement as detrimental to employment prospects.

The theories of innovation are a historically rich field of research. One guiding theory is the *Combinatorial view of Innovation*, originally envisioned by Schumpeter [5]. This is the view that much of innovation derives from combining aspects of existing knowledge in novel ways, and it is empirically justified in a wide variety of scenarios.

For example, the widely cited work [6] examines the nature of innovative research papers, and finds “The highest-impact science is primarily grounded in exceptionally conventional combinations of prior work yet simultaneously features an intrusion of unusual combinations”.

Rogers’ (1983) *Diffusion of Innovations* is another theory of innovation focuses on the factors that determine how diffusive an innovation is - diffusion in this case is the “process by which an innovation is communicated through certain channels over a period of time among members of a particular culture” [7]. Rogers depicts that there are four elements affecting the spread of new ideas, which model the transition as resulting from social and network factors. The wide-adoption of an innovation is integral to its sustenance on this view, but its key insight is that social networks hold the key to understanding whether an innovation will prevail.

There are many other theories of innovation. The authors of [8] assess and classify the meta-theories of technological innovation from classical texts and modern sources. Among their findings is that there are a wide variety of views on the dynamics of technological innovation, each exhibiting different causal understandings as depicted by the authors’ meta-theory classifications.

2.1.3 Role of Artificial Intelligence

Artificial intelligence is perhaps in a separate category of technological innovation for a few reasons. As is its nature, it is a different type of technology as it is

inherently regarded as *intelligence*. This can be seen as both complementary and substitutional to human intelligence or labour [1].

This gives rise to emergent phenomena as it relates to technological innovation and labour:

1. **AI *as* innovation:** AI as a technology in the classical efficiency multiplier sense.
2. **AI *drives* innovation:** The authors of [9] examine the effects that AI has *on other* technological innovation, finding a strong positive relationship, especially in low-tech sectors.
3. **AI *replaces* innovation:** Artificial General Intelligence (AGI) is the abstract concept of a technology with an intelligence surpassing that of all humanity [1].

It is the 3rd point here that is of deep importance, because it represents such a profound shift a civilizational sense. Aside from the controversy and disagreement that surrounds the topic of AGI [1], it is clear that Labour demand will be profoundly impacted by the continual development of AI [1].

2.2 Labour Demand Forecasting

2.2.1 Overview

The practice of forecasting the demand (and *supply*) for labour has a long history [1]. Many critique its utility [10], but there is good reason to believe that the limitations faced by current approaches are not insurmountable. What follows is the findings of research into existing approaches, with an emphasis on their limitations.

2.2.2 Typological Approaches to Forecasting

2.2.2.1 Manpower Requirements Approach [10]

The authors of [10] discuss the *Manpower Requirements Approach (MRA)* for forecasting labour demand, within the context of Aboriginal Canadian communities. For context, the MRA approach is still central to the US's Bureau of Labour Statistics (BLS) occupational-forecasting methodology, among other major institutions. In summary, the approach utilizes a “future macroeconomic reference scenario for GDP [...] to derive an estimate of total employment demand, which is later broken down by industry and by occupation.”

The authors modify the approach slightly, due to some concerns about feasibly producing “macroeconomic reference” scenarios as they concern the Aboriginal population of Canada. As a result, alongside a separate model they define for labour supply, their methodology is considered to be a fairly comprehensive framework for understanding demand & supply.

An important contribution they make is to lament the importance of forecasting not only “Expansion” demand (an increase in demand in absolute terms) but also “*Replacement*” demand (an increase in demand resulting from gaps created by mortality, retirements, occupational mobility, etc.).

However, the methodology is deeply assumption-led and is consequently limited.

- **Static occupational coefficients:** They assume that occupational distribution by industry remains constant over time
- **Zero elasticity of substitution between labour types:** workers in different occupations aren’t easily substitutable.
- **Extrapolation of trends:** e.g., labour force participation or enrolment rates follow past patterns—even if future social or structural shifts may alter them significantly.

Furthermore, as the authors recognize, the framework is fraught with other issues including:

- Data Limitation problems for smaller institutions
- An assumed heterogeneity of individuals within occupations misses a granular understanding of the individuals’ differences (e.g. their skill-sets)

To add to data limitation problems, the authors find that a best practice is to use “up-to-date data and regularly refresh forecasts (every 2–3 years, medium-term horizon of 5–10 years)”. This is insufficient with respect to today’s economy, especially considering the fast-paced shift in the labour market as a result of AI.

This review of labour forecasting methods is not without its issues, but highlights some important considerations for future endeavours.

The major limitations of their approach are:

1. Lacking granularity beyond occupation-level
2. The assumptions are highly simplified
3. Data requirements are extensive
4. Time-wise granularity is poor

Moreover, occupation-level findings of such forecasts are increasingly irrelevant, for reasons detailed later.

2.2.2.2 Quantitative Econometric Approaches [11]

A still prevalent and perhaps predominant approach to demand-side forecasting is through econometric macroeconomic models: linking macro indicators (like GNP or industry-level output) to employment and occupational demand. These are often regarded as theoretically sound because of how they respect the underlying economic principles regarding individual behaviour. To a large extent, these models are supremely effective in modelling large-scale phenomena resulting from the microscale (e.g. individual/household level). A contributor to their successful usage is their capacity to express causality resulting from a vast array of factors (wages, capital, trade, productivity, technology, etc.). In any model of a real-world system, simplifying assumptions are necessarily present and as such, the major concern is that they do not represent the real-world sufficiently well - it is debatable that this even matters if a model is predictively powerful [12].

But there are major limitations to solely depending on econometric models:

- **Time Horizon:** Though purporting to estimate with long horizons, they often break down after the medium-term.
- **Data problems:** Expressive econometric models require long-running detailed time-series of various data factors. This is demanding even for economies with developed data infrastructure, and simply unfeasible for those with less developed data infrastructures.
- **Limits to understanding:** across time, parameters may evolve to be more/less relevant, or their relationships alter fundamentally - this change is not accounted for by the top-down specification of the model assumptions.
- **Fallacy of past behaviour being indicative:** problematic in the face of widespread technological disruption as is expected with AI.
- **Technical (Statistical) difficulties:** often, highly technical models are not useful unless they are highly interpretable, which remains difficult for non-technical policymakers or employers.
- **Inherently unpredictable events:** the models often suffer / break-down during periods of economic shock (e.g. financial crises, technological shifts).

In practice, such models are often best used in combination with other approaches, such as survey-based or other empirically grounded techniques. Furthermore, they discuss and address explicitly that though approaches like MRA utilize an occupation-level granularity for forecasts, the lost nuance is detrimental to their defensibility. They recognize that policy-makers and governments will find

two/three-dimensional forecasts (relating to industry/occupation/qualification dimensions) increasingly irrelevant and incapable of producing informative forecasts in the face of those utilizing multidimensional data (“functions, tasks, abilities, personal attributes, characteristics and generic skills”).

Data Limitations and detailed understanding are the primary concerns that the authors raise as to the viability of econometric forecasts.

2.2.2.3 Survey-based Approaches [13]

Another common method of gauging labour demand is through employer or establishment surveys. These typically involve asking firms directly about current vacancies, expected future hiring, or perceived skills shortages. They can be tailored to particular industries or occupations, offering a more immediate and micro-level picture of demand.

The strength of survey-based approaches is that they can capture employer expectations in near real-time, and may reveal demand for skills that are not well reflected in historical data or official occupational classifications. They are also relatively straightforward to implement and can provide insights into qualitative aspects of demand (e.g. “soft skills” requirements).

However, their limitations are notable:

- **Short time horizons:** employer expectations rarely extend beyond 1–2 years, limiting their usefulness for medium- or long-term forecasting.
- **Sampling and response bias:** firms willing to respond may not be representative, and self-reported expectations are subject to optimism or pessimism.
- **Lack of standardization:** surveys are often ad hoc, making results difficult to compare across sectors or countries.

These issues mean that survey-based forecasts are best viewed as complementary, rather than substitutes, for model-based approaches.

2.2.2.4 Input–Output Approaches [11], [14]

Input–Output (I–O) approaches use inter-industry transaction tables to model the relationship between output in one sector and the demand for inputs (including labour) in others. By combining I–O tables with projected sectoral growth, forecasts can be generated for labour demand by industry and, through further assumptions, by occupation.

This approach has the advantage of providing a structured, economy-wide framework that captures interdependencies between sectors. It is particularly useful for analysing the employment effects of shocks or structural changes in specific industries.

However, several limitations arise:

- **Static technical coefficients:** I–O models generally assume fixed relationships between inputs and outputs, ignoring substitution and technological change.
- **Data intensity:** compiling detailed and up-to-date I–O tables is resource-heavy and not always feasible for smaller economies or specialized subgroups.
- **Poor skill granularity:** I–O models focus primarily on industry demand, meaning occupational and skill-level forecasts require additional strong assumptions.

Consequently, while valuable as part of a comprehensive forecasting toolkit, I–O approaches suffer from the same rigidity and assumption-dependence as the manpower requirements and econometric methods discussed earlier.

2.2.2.5 UK Context

Robert A. Wilson, the author of [11], contributes also to many specific forecasting efforts especially considering the UK government’s Working Futures programme (WFP). This is the primary tool that the UK government uses projecting future employment demand, and is very comprehensive.

The Working Futures Report [15], produced by the Institute for Employment Research (IER) of Warwick University [16] in partnership with other institutions (including the UK Commission for Employment and Skills (UKCES) and the Department for Education), is a “comprehensive, systematic, consistent and transparent set of projections” of the likely future’s labour market. Their projections are centrally grounded by projections of the E3ME Cambridge Econometric Model [17], producing multi-sectoral & multi-regional projections for 75 industries. Their rationale suggests that “sectors / industries [are] the key driver of changes in employment by occupation” [15]. Producing 10 year projections at intervals of 3 years, the report also makes use of with modules for replacement demand, occupations, and qualifications, with data from the Office for National Statistics (ONS) [18]. Framed not as exact predictions, the purpose of this forecast is to inform policy given outlooks or views of potential futures.

An immediately noticeable limitation of this approach is the periodicity of updates. For example the 2017 – 2027 forecast is produced in 2020. With a base

year of 2017, the forecast can only be produced after a lag-time for the purpose of acquiring the relevant data, such as national figures, which are released at a lag of 1 – 2 years. Thus, the next report, grounded in 2020, would be expected to release in 2023 - this most recent report is structured somewhat differently, with a longer time-horizon (2020 – 2035) [19].

To some extent, the report is immune to a critique of its long-term accuracy, given its framing as a “future likelihood” tool. Nevertheless, *improved* accuracy is always tangibly beneficial. Those critiques (as the report itself details) discuss those limitations:

- The forecast is heavily assumption-led
- The forecast results are not designed for precise local-area forecasts
- The forecast may lack the nuance of *intra*-occupational change

Further, a critique produced by [20] finds that many forecasts share the same limitations:

- Difficulty in linking from macroeconomic trends or employment forecasts to skills
- Few attempt to understand how skills change *within* occupations.

2.2.3 Alternative approaches

There are a number of solutions that are being investigated that aim to resolve some of the aforementioned limitations of labour market forecasting.

In the past, “naturally occurring data” (data not produced for analytical purposes) such as online job postings were difficult to assess systematically because of their lack of structure [21]. Furthermore, it has not historically been the case that vacancies are always advertised online, thereby adversely affecting their strength in representing labour market demand. However, these concerns are both increasingly irrelevant, due to advancements in Machine Learning techniques and the almost ubiquitous practice of advertising vacancies online.

These offer a potentially powerful data-source for understanding how the labour market is changing, and are under thorough investigation in many situations. One example is how the author of [22] uses Job Posting text to predict salaries, demonstrating the informative and predictive power of the information expressed by the natural text source.

Among those most pertinent to this paper are those that extract and portray the evolution of skill-requirements.

The Bank of England (BoE) employ an unsupervised Machine Learning (ML) techniques with job-posting data to identify disaggregation/segmentation methods for forecasts that are not at a sector/occupation level as in the WF Report, but are instead at a skills-level [23]. They recognize the power of this Machine Learning based approach to devise data-driven taxonomies for skills that are otherwise unobserved by official classifications, which “cut across wage, sector and occupation”.

The European Centre for the Development of Vocational Training (CEDEFOP) have an ongoing project in their *Skills Online Vacancy Analysis Tool for Europe* (Skills OVATE), which uses vast data pipelines from 27 Nations (EU member states, EEA countries and the UK) to produce *quarterly* updates on a skill-demand forecast [24]. Another example of how skills can be investigated using job postings is presented by the OECD. They analyse Job-posting to produce an outlook on the demand for AI-related skills across sectors [25].

Other research suggests the power that Job Postings have to produce meaningful insights into how skills within occupations are changing [26]–[28].

2.2.4 Key Insights

The approaches available for a given economic actor to produce forecasts of labour market demand are varied. Whilst mostly different in their these methods, there are shared problems.

In general, for comprehensive models, high quality & voluminous data are needed at regular intervals, from a panoply of sources. This is deeply problematic not only for those private individuals with few resources available to them, but also to firms and even policy-makers, since acquiring such data introduces a necessarily longer lag-time for new forecasts. Furthermore, macroeconomic models experience difficulty in projecting intra-occupational change - this is fatal for properly understanding a labour market which is driven fundamentally by the more granular unit of individual skills, from which occupation-level requirements are composed.

As such, a variety of solutions are being investigated that take a more data-driven and bottom-up approach for understanding this more functional disaggregation of labour-demand - i.e. in terms of *skills*. Integrating the information observed in naturally-occurring data such as job postings to aid forecasts is ongoing, but they nevertheless show promise due to the informative, high-frequency and high-volume nature of job posting data, and are therefore worthy of consideration.

2.3 Open Source Software (OSS)

2.3.1 Open-source concept

Open-sourcing is a powerful concept within technological information. Before the internet, published information was centrally controlled, by the publishers or curators of such information. With the conception of internet tech, widespread collaboration on such information sources became viable, leading to openly available and extensively developed sources of information such as Wikipedia - this version of an encyclopedia is different to its paper-printed ancestors because *anyone* has the ability to propose and implement changes (under administrative audit). As a result, Wikipedia is the go-to encyclopedic source of information, known for its vast and detailed coverage of topics, as well as its reliability in presenting up-to-date and trustworthy information.

2.3.2 Open-source for Software

Open-source software (OSS) is the release of software under a licence that permits (under varying conditions) the software's modification and redistribution [29]. On the whole, OSS is designed to promote collaboration and innovation within software, and is a paradigm under which many of the most powerful software tools are developed. Analogous to what Wikipedia is for general information, the most well known "host" of OSS is GitHub - this is a platform which provides extensive access to, and methods to collaborate with (or contribute to), OSS projects. There are other such platforms, but GitHub is by far the most popular. Central to what makes GitHub successful is its powerful version-control system (Git) - functionally, this is the mechanism by which multiple people can work on a document simultaneously without conflicting changes, as well as providing routes to revert changes and control permissions.

For the remainder of this paper, I shall use interchangeably the notions of OSS and GitHub:

Stipulation

The terms **GitHub (GH)** and **Open-Source Software (OSS) Platform** (and equivalently, **GH repositories** and **OSS products/projects**) are used interchangeably.

Though GitHub is not the *only* source-code host for OSS, it is by far the most popular [30]. Furthermore, given that the scope of this discussion is not that of GitHub (the platform) and instead concerns its hosted projects (i.e. repositories), this stipulation is relevant because I intend to use GH repositories as proxies for

OSS projects in general (The defensibility of this proxy assumption is examined under Section 5.1.1). This is an important distinction, and promotes pragmatic/-effective discussion without unnecessary jargon.

2.3.3 OSS Importance

Many view that open-source is a practice that underpins the success and advancement of software technology. As the authors of [31] report, OSS should be treated as a public good. They further lament the unseen cost to developers within OSS, as well as specifying the necessity of upkeep for such ecosystems.

Indeed, given the expectations transformative power of AI, it is regarded open-sourcing relevant software is more important than ever [32], [33]. It is seen as vital that powerful software systems such as AI ought to be democratized to prevent exploitation arising from private ownership, allowing the benefit to distribute more evenly among the population: “We believe the open source approach is the only way to achieve the full potential of AI, making it safer, accessible and democratized” [32].

2.3.4 OSS as Innovation

Open source software (OSS) is increasingly a rich ecosystem for The development of new software technologies. Often it is regarded as a hub for experimentation and innovation, and is a unique depiction of innovation outside the incentive structure observed by private economic agents.

Weak ties (low-intensity, infrequent interactions) are a long-theorized and recognized driver of innovation, since in a broad sense, they expose individuals to ideas from unrelated economic & social circles [8]. The authors of [34] investigate the impact of weak-ties, in the context of software development circles, on software innovation. Their findings suggest that a developer/contributor’s future project innovativeness is strongly correlated with the breadth and diversity of projects they engage with. This is highly suggestive of the power that the OSS ecosystem has to induce a positive externality-like effect for software innovation.

Similarly, the authors [35] examine “the imports of libraries in 12 different programming language ecosystems within millions of Stack Overflow posts over a 15-year period”. They aim to present learnings of OSS innovation to the literature on innovation dynamics, and find that recombination is a key innovation process in software (resembling concepts such as combinatorial innovation [5]). This is novel because they demonstrate the efficacy of treating the OSS ecosystem as similar to patenting and scientific research in terms of the innovation dynamics they express. The authors of [36] observe similar dynamical traits, doing so with

a novel approach of examining the dependency growth and complexity of GitHub repository.

2.3.5 Quantifying OSS innovation

A key consideration for understanding the impact or proliferation of an OSS project, thereby in attempt to measure its innovative qualities, is defining how this is *measured*. GitHub, being designed for maximally efficient & transparent collaboration with optimal version control, naturally offers openly available metrics for projects as relating to their development. These include *forks*, *commits*, *contributors* and so on. Within each there will be detail regarding the specific instance, the time and their specific natures.

For example, a commit will include information on what files have been changed and a commit message describing those changes. A *contributor* is a user who has, at least once, made a commit to a project - a contributor-project association graph can thus be defined. A *fork* is a duplication of a project in its current state to a separate *branch*, useful for refactoring or adding new features without affecting the working (production) source code. When satisfied with the outcome of a fork, a merge is created to update the main branch.

These explanations miss a little nuance, but it remains clear that there are several avenues for tracking and monitoring the development of a GitHub repository.

One other metric, that is somewhat seen separately to those naturally occurring data regarding a repository’s development, are “GitHub stars”. The literature regarding stars suggests a lack of clarity as to their primary purpose, mostly because there isn’t a functional one as regards development.

A major study into the usage and impact/relevance of stars is conducted by the authors of [37]. In earlier work they investigate the popularity growth of GitHub repository [38]. They “[identify] four main patterns of popularity growth, which are derived after clustering the time series representing the number of stars of 2,279 popular GitHub repository”. Among their findings is that stars & forks are strongly correlated, but Stars & contributors and stars & commits are weakly correlated. Additionally, they find that repositories with high number of stars are overwhelmingly owned by companies or institutions, rather than individuals. A key insight of theirs is to suggest that many repositories see a burst of star-gains after their initial release, which almost always drop off immediately after - this suggests that bursts of popularity are not indicative of long run popularity growth.

A common theme of their research is to compare the popularity of GitHub repository to similar studies in social media popularity phenomena, such as classifying popularity growth patterns: Slow, Moderate, Fast, and Viral.

Expanding in later work [37], they:

1. Expand their analysis to 5000 repositories,
2. Survey 791 developers to identify the motivation / reasons behind starring,
3. Investigate of factors influencing star-growth (e.g. marketing),
4. Develop a Machine Learning method to identify specific factors that predict a GH repository’s popularity growth.

Among their primary findings is that “Stars are a key metric about the evolution of GitHub projects”, and as such their monitoring is important for project managers. They identify that, in addition, what prompts *fast* star growth is often the result of “active promotion, usage of trending technologies and innovative projects”. However, *virality* is presented as a caveat to the general treatment of stars as a repository’s popularity: often, this is the result of marketing and social media promotion. As such, the high star-count is often a result of a burst of stars gained during certain periods. They therefore urge caution when selecting GitHub projects by number of stars, and to check whether these stars were gained over sustained periods or within short time-frames.

In [39], the relationship of a repository’s stars and its downloads are investigated. In summary, their results find that both are “indicators for the importance of client-side JavaScript libraries”. Whilst not immediately relevant to all types of GitHub projects, such as those in other programming languages (e.g. Python) or for other end-goals (such as for AI or ML), the results are nonetheless promising in support of Stars as a metric for project importance.

The authors of [40], building on the research supported theoretical concept of combinatorial innovation, examine the trends and relationships of such recombination projects in the Python programming language. Their results suggest that higher levels of innovativeness are strongly correlated with high-star counts - “Novelty begets popularity”. Interestingly, they also note that the relationship between contributor-count and innovativeness is inversely correlated.

An interesting approach followed by [41] attempts to build a *predictor* of repository stars given other features of a repository, using a few Machine Learning models. The features they use fall within categories of:

- **General Project Characteristics:** locality, commits per developer, number of forks, number of pull requests (PRs), total contributors, repository age, programming language (and more)
- **Characteristics of Developers and organization:** followers per developer, developers per organization (and more)

- **Activity:** commits per day, ratio of accepted/total PRs

Their best results arose from using PCR and Random Forest predictors, with R^2 (explained variance in the dependent - i.e. stars - as a result of the independent variables) values of 0.87 and 0.88 respectively. With an analysis on how the mean squared error (MSE) of the predictions change with the removal of features, they find that the most important features are:

- Number of forks
- Number of followers per organization member
- Number of PRs
- Number Average number of commits per day
- Maximum number of followers per organization member

The most prominent of these (forks) is strongly supported elsewhere, but otherwise, the “follower” trait of an organization member is a surprising and novel result.

This suggests that popularity is observable not only in stars, nor only in forks, but also in the network characteristics of a repository’s contributor community.

2.3.6 Key Insights

Within software, OSS is much more relevant in understanding innovation than are other innovation signals (such as research papers or patenting). Whilst there is perhaps strong evidence to suggest that multiple metrics would form better appreciation for the innovativeness and utility of a GitHub repository, there is strong support for treating stars as a proxy measure for popularity. By extension, since popularity is a good indicator of novelty, stars are a simplistic but often effective measure of a repository’s innovativeness but with caveats: the timeline of stars is of equal importance, since bursts of star gains are not indicative of much more than strong promotional and/or marketing efforts. Instead, long sustained star-growth trends are more indicative of innovativeness. Moreover, it is worth noting that under Rogers’ [7] theory of innovation, it is diffusion through networks that results in the overall success of an idea - this should be seen as a supporting principle of stars as a measure of innovation diffusion, at least within the software community. Furthermore, the weak-ties investigation [34] is somewhat analogous to [6], and supports this idea that OSS ecosystems exhibits Schumpeterian tenets too. It’s nice to see (unrelated) classic economic theories have something informative to say about OSS ecosystem’s innovation system.

2.4 Natural Language Processing

2.4.1 History

The simplest Natural Language Processing (NLP) technique is keyword matching: querying a body of text to determine the presence of a defined keyword. Though straightforward, This method is too narrow to capture the complexity of *natural* language, which encompasses alternative terms of phrase, acronyms and so on. To be useful, this method would involve curating a large dictionary of terminology, which is laborious.

An improvement would be to use fuzzy matching techniques, which allows for matching on typos and some small patterns alternative phrasing. However, this approach remains limited in its ability to capture *meaning*. It is not enough to simply define a match on a word-for-word basis, because a crucial element of what makes natural language meaningful is the context. For example the word “Python” alone means something very different when seen in the context of “Python, cobra and anaconda” as compared to “Python, SQL and Java”.

A major breakthrough in NLP came with the shift from sparse, dictionary-based features (like standalone words or phrases) to *dense vector embeddings*, where words and phrases are represented as points in a continuous high-dimensional space. Early methods such as Word2Vec [42] and GloVe [43] captured semantic similarity by training embeddings such that words occurring in similar contexts are placed near one another.

The subsequent development of *self-attention* mechanisms, formalized in the Transformer architecture as depicted by the groundbreaking *Attention is all you need* paper [44], extended this idea by dynamically weighting the relevance of surrounding tokens when generating a representation. This allowed embeddings to become *contextual*: the same word, such as “Python”, receives different vector representations depending on its sentence context (programming vs. reptile). Together, distributed embeddings and self-attention form the foundation for modern models like BERT, enabling scalable and context-sensitive language understanding.

2.4.2 BERT and Sentence Embeddings

BERT (Bidirectional Encoder Representations from Transformers) [45] is trained using a masked language modelling (MLM) objective (in which tokens in a sentence are randomly masked, and the model must predict them using the surrounding context) and a next sentence prediction (NSP) objective (which trains the model to understand sentence-to-sentence relationships). Together, these tasks

yield rich token-level representations that are *context-dependent*: the embedding of “Python” differs depending on whether it appears alongside other programming languages or reptiles.

In practice, BERT encodes each token in a sentence as a dense vector in a high-dimensional space. Because of the bidirectional self-attention mechanism, these token embeddings capture both syntactic and semantic relationships with their surrounding context. Downstream NLP systems can fine-tune BERT’s parameters, or simply use its representations as input features, to achieve state-of-the-art performance on a variety of tasks (e.g. question answering, named entity recognition, or classification).

However, BERT is primarily designed to produce *token-level embeddings*. When a sentence/document-level representation is required, further processing is needed. A common approach is to average the embeddings of all tokens, or to use the special [CLS] token, which is trained to aggregate sentence information. Later models, such as Sentence-BERT (SBERT) [46], build directly on BERT to optimize for high-quality sentence embeddings by introducing a contrastive training objective. This allows semantically similar sentences (e.g. “She enjoys programming in Python” and “She likes coding in Python”) to be mapped to nearby points in the embedding space, while dissimilar sentences are mapped further apart. Sentence embeddings thus form the basis of semantic similarity search, clustering, and retrieval-augmented methods that are critical to modern NLP pipelines.

2.4.3 Natural Language in Job Postings

2.4.3.1 Skill and Competence Taxonomies

To represent the “skill profile” of a Job Posting (JP) or a GitHub repository (GHR), we need a **taxonomy** to define the concepts of a “skill”. Official classification bodies such as ESCO and O*NET exist for this broad purpose. They define hierarchical classifications to depict skills requirements for occupations.

The term *skill* itself is, however, only narrowly defined by the *European Skills, Competences, Qualifications and Occupations* (ESCO) classification [47] as “the ability to apply knowledge and use know-how to complete tasks and solve problems.” ESCO further distinguishes *knowledge* as “the outcome of the assimilation of information through learning,” encompassing facts, principles, theories, and practices related to a field of work or study, and defines *competences* as “the proven ability to use knowledge, skills, and personal, social, and/or methodological abilities in work or study situations and in professional and personal development.” Thus, competences capture the broader capacity to apply both knowledge and skills within specific contexts.

These classifications are problematic for the purposes of this project, because in practice, ESCO does not make a clear distinction between skills and competences within its “skills pillar”. ESCO also defines many items that are ability-wise relevant to AI (e.g. specific programming languages) as knowledge rather than skills. Additionally, ESCO classifications are manually defined, and may therefore be missing the nuance of recent & emergent taxonomies in the domain of AI. The strict and somewhat outdated ESCO classifications are therefore problematic for framing what the “skills” in this domain are.

Therefore, for the remainder of this paper, I shall stipulate the following:

Stipulation

The term “skill” is in this paper used broadly to encompass all three ESCO categories insofar as they relate specifically to AI technology.

This serves to simplify the discussion of how GitHub and Job Postings relate within the domain of AI technology, and should therefore not be confused with how ESCO defines a skill.

2.4.3.2 NLP for Skill-extraction

In [48], the authors develop a data driven methodology to extract and represent a job-skill taxonomy. Emphasizing that they do so independently of official boards for taxonomy definition (e.g. O*NET and ESCO), they Leverage NLP and network analysis methods for an algorithmic methodology to construct skill taxonomies, which remain adaptable and avoid expert elicitation. The significance of this is that it reinforces the feasibility of using NLP techniques, and specifically sentence embeddings, to extract skills from job postings.

Named Entity Recognition (NER) is a method of Natural Language Processing which identifies and extracts known entities (such as companies, people, or locations) from a body of text. The authors of [49] develop a specialized annotated dataset of 301 Job Postings (JPs) that serves to validate NER models for skills & knowledge (treated separately) extraction. Their annotations extend beyond the basic token-wise annotation as regularly done with NER models, and instead use *span* labels (using BIO (Beginning, Inside, Outside) tagging). This is done because often, skill-entities are extended spans of text (e.g. "Ability to debug code").

Their process takes 2 baseline models, *BERT* and *SpanBERT*. They utilize a set of unlabelled 3.2M Job Postings to domain-adaptively pre-train these two models to produce a further 2 models, which they call *JobBERT* and *JobSpanBERT*.

With their annotated dataset, they fine-tune each model by passing sentences

through the model’s transformer encoder, through a linear layer, after which the output of a Conditional Random Field (CRF) (“framework for building probabilistic models to segment and label sequence data” [50]) layer returns BIO-tags for the sentence. The models are trained to maximize sequence-level likelihood under the CRF.

In essence, their research contributes:

1. A novel dataset of labelled Job Postings with span-level annotations: span-level annotations are useful because they are more layered than basic token-level annotation. E.g. They allow for nested tagging: “ability to use Python programming libraries” - as a whole span of text, this is seen (as under ESCO classification) as a *skill*, whereas Python is itself considered *knowledge*. Both aspects are tagged within their specialized dataset.
2. Evaluate the performance of baseline vs. domain-adapted BERT and SpanBERT models on this annotated dataset.

They find that their domain-adaptive pre-trained models outperform baselines, with JobSpanBERT performing best on Skills and JobBERT performing best on knowledge. The results of this paper are such that there is purportedly clear benefits to adaptively training a BERT, and hence transformer-derived, models on domain-specific data, but this is less true regarding knowledge components: a difference of 1.5 F1 for the BERT and JobBERT on knowledge.

A limitation of this paper is the small dataset of annotated examples by which they use to evaluate their models’ performances. As is often the case, data annotation remains a barrier to extensive and nuanced models capable of generalized high-performance in this case.

2.4.4 Natural Language in GitHub repository

A growing body of work demonstrates that the natural language fields of GitHub repository—such as titles, descriptions, README files, and user-defined topics—encode rich signals about their technical scope and knowledge content.

Repo2Vec [51] combines textual metadata with structural and code-based information to create repository embeddings, showing that natural language fields alone were sufficient for effective similarity search, clustering, and even malware detection.

Another function is to utilize NLP methods for building *topic recommendation* systems. A feature of GitHub repository is the option to tag a project with relevant topics - these are useful for navigating between and grouping similar repositories. However, repository owners do not always tag their projects sufficiently, and often

not at all. Solutions to this treat repositories as multi-label text classification problems, concatenating descriptions, READMEs, and other textual components, and applying both traditional document representations (TF-IDF, Doc2Vec) and contextual embeddings from Transformer models such as DistilBERT [52]. These methods achieve strong performance in predicting repository tags and demonstrate that semantic representations of repository text align well with human-curated technical taxonomies.

Other work extends this principle by linking repositories to external knowledge domains. For example, *paper2repo* have been developed to match academic papers with relevant repositories by embedding both corpora in a shared semantic space [53]. The success of this method is important in that it justifies, to some extent, the viability using semantic embeddings for comparing the similarity of textual content *across domains*.

Graph-based approaches such as HiGitClass [54] similarly leverage README and description text as node features in heterogeneous networks, showing that textual content improves repository classification across languages and domains. More recent efforts use encoder-only LLMs, such as BERT, DistilBERT and RoBERTa, to produce Transformer-based sentence embeddings that classify README sections into functional categories with high accuracy [55], further underlining the representational value of repository documentation.

Taken together, this line of research supports the view that GitHub repository can be modelled as vectors of their textual features. These embeddings capture the *technical concepts* (e.g. programming languages, libraries, frameworks) and *knowledge concepts* (e.g. problem domains, tasks, skills) embedded in repository descriptions and documentation, making them a valuable feature source for downstream analyses of technological innovation and labour demand.

2.5 Synthesis and Research Gap

2.5.1 Integrated Understanding

On the whole, then, this extensive literature phase has informed the following takeaways:

1. Comprehensive labour demand forecasting methods are time-intensive, data-heavy and struggle to capture nuances of skill-level changes. They are laden with assumptions and are not designed in a manner that expects underlying relationships between factors to change. Given the fast-paced and expected highly disruptive nature of AI development, there is great need for

time-effective and data-driven forecasting methods that are adaptable and relevant.

2. Job postings present an avenue for such data-driven forecasts. Given their nature as high-frequency and high-volume data-points, they can reliably be treated as proxies for real-time labour demand. While historically they have been unfeasible to mine and represent in an analytically useful manner, new methods of NLP overcome this limitation.
3. OSS and GitHub repository are widely viewed as key to innovation in software technology, much of which AI and AI-related competency is connected to. The popularity of such OSS projects on GitHub (namely, repositories) can be, to some degree, measured by a proxy-measure of their “stars”. This is an increasingly recognized manner in which the conceptual themes of technological can be assessed: by monitoring the stable, and continued, growth of stars in a repository.
4. GitHub repository can be represented as a feature of their textual content, breaking into topical/technical classifications that extend beyond their explicitly defined topics. Furthermore, there is some evidence to suggest the validity of producing cross-domain embeddings that can be assessed in a semantic context, such as via similarity. This is supportive of an approach to represent GitHub repository in terms of features that are shared with other textual corpora (e.g. Academic papers, as shown, and potentially Job Postings).

There is not, as I am aware, a concerted research effort being conducted into assessing the **spillover** of innovation from OSS $\rightarrow$ Labour Demand. This is a major gap in current research and needs investigating.

2.5.2 Justification of Current Approach

The methodology of this paper therefore aims to explore this gap in understanding. With my understanding of the *Diffusion of Innovations*, those that have pervasive and sustained impact through social circles are the innovations with the highest impact. Therefore, it stands to reason that such strong innovations in *software* (whose most innovative catalyst is that of OSS ecosystems) are resultant in later demand for *knowledge* of that software.

This cross-domain understanding is then assessed based on the quantifiable existence of:

1. Temporal Job Posting as proxy labour demand

2. Temporal GitHub repository stars as popularity/diffusion

Additionally, NLP methods are supportive of:

1. Representing Job Postings as expressive of skills
2. Representing GitHub repository text in terms of technical details

This research aims to develop a cross-domain representation pipeline for the textual elements of Job Postings & GitHub repository, thereafter utilizing their respective temporal modalities that are useful for answering the diffusion question.

Chapter 3

Methodology

3.1 Chapter Introduction

The purpose of this chapter is to more concretely depict the stages of the pipeline detail what is involved, step by step. My usage of the word “pipeline” is inspired by the nature of the implementation being a full, end-to-end solution (as opposed to specific instances or model developments). A crucial part of what makes such a pipeline successful is a reliable and operational *flow* of computational tasks.

In a brief sense, we can suggest that there are three main stages to this pipeline, all of which are in terms of underlying source(s) of data:

1. **Acquisition:** finding appropriate data, and preparing it for “processing” readiness.
2. **Processing:** Otherwise called “representation”, this step involves transformations of the acquired data to become analytically useful. In ML, for example, this is the training of a model (using the acquired data) to produce meaningful outputs (predictions/classification) that may thereafter be used for analytical purposes.
3. **Analytics:** Investigation of underlying data, as represented by the intermediate form, with regards to the initial research question.

Whilst vague, this can help guide expectations for the various steps in between, since with long and complex pipelines it can be difficult to assert the full relevance of each stage in isolation. A version of this pipeline, therefore, is represented neatly by the diagram in Figure 3.1.

Specifically then, if the goal is to measure diffusion of OSS innovation into labor demand, then We want to measure the relationship between two temporal series:

1. The trend of *innovation* - The evolution & popularity on the development side,

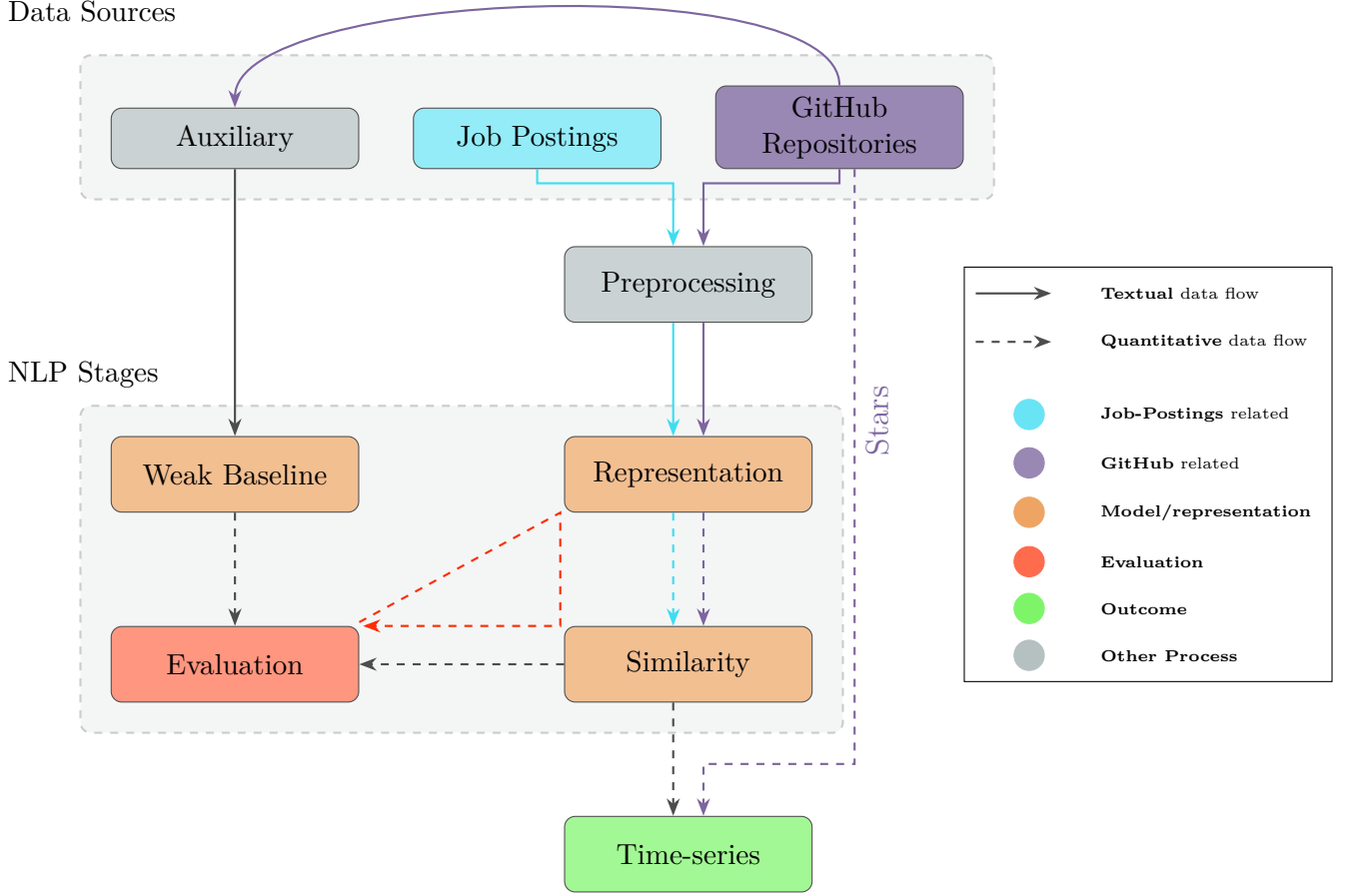


Figure 3.1: Conceptual diagram of methodology pipeline.

2. and the trend of *adoption* - The requirement of possessing the relevant knowledge/skills.

The first of these is straightforwardly representable by the so-called supply side of development (specifically, GitHub) metrics. The second, however, is challenging: there is no straightforward way to depict whether a repository is relevant to a job and vice versa.

Therefore, along the way, we must construct a method to represent this relevance factor. As the literature suggests, **Sentence Embeddings** present a clean solution - in an intra-domain sense, the method is demonstrably viable for evaluating the semantic similarity of documents. In addition, there is evidence to suggest that *cross-domain* semantic similarity can be assessed using sentence embeddings, though there is no research currently that supports this idea in this context (Job Postings vs. GitHub repository).

The bulk of this paper is therefore dedicated to establishing a **high quality representation of cross-domain semantic similarity**. The following describes each component of the pipeline in detail:

1. Data collection,
2. ETL & EDA (Pre-processing, Exploring & Cleaning),
3. Weak model development (Fuzzy-matching keywords),
4. Sentence embedding representation,
5. Evaluations and experimentations (Textual Variations, Sentence-transformer variation, Dimensionality Reduction),
6. and Alignment & Analysis (Correlation, lead-lag) of resulting time-series’.

3.2 Data Collection

This phase of the project establishes a systemized workflow to find and pull data. This effort can be grouped in three distinct ways categories (or channels):

1. Job Postings,
2. GitHub repository,
3. and Auxiliary data.

3.2.1 Job Postings

3.2.1.1 Overview of Data Landscape

As previously discussed, the practice of online advertisement of employment vacancies is a ubiquitous in recent times. In one sense, this is a benefit because concerns about existence of data are not pertinent. A panoply of websites, colloquially known as “job boards”, exist online - the primary examples of these include *LinkedIn* [56] and *Indeed* [57], though these sites are much more than vacancy-advert sites.

Some job boards have dedicated API layers. For example, among the various available LinkedIn API endpoints (under paid access) is one that allows for efficient programmatic interfacing to find vacancies in occupational-fields or specific roles - this is useful for both employers and employees alike. In other cases, web-scraping is a viable option, albeit more laborious than using APIs.

3.2.1.2 Problems in Acquisition

However, a severe limitation of acquiring data in real-time from these sites is a result of the fleeting nature of Job Postings: Once a vacancy is filled, the advert is no longer necessary, so the data-point is no longer maintained on the job board site. Because of this fact, directly accessing Job Posting data from job boards means retrieving that data as it appears (and before it disappears) in real time.

Given that the scope of this paper investigates the long-term relationship between data-sources (diffusion can be a slow process), it is not enough to *begin* collecting data now.

Rather, it is necessary to acquire pre-existing datasets consisting of historic records. This proved problematic, because largely, such data is expensive to store, and as such is not considered by job-board sites as necessary to maintain. Moreover, because of this economic cost of storing this data, the most comprehensive access to such data is expensive: Firms such as Lightcast [58] are framed as “labour market intelligence” firms that offer data & analytics services, pertaining to Job Postings under the labor market data.

3.2.1.3 Solution: Kaggle

Thankfully, the open data community of Kaggle [59] proves mightily useful for such data needs. This is a platform for Data Science and Machine Learning where people share lessons, data & code for education and competitive purposes. As a resource, it is often the solution to data acquisition: in most cases of niche requirements, someone has likely created a relevant dataset on Kaggle.

It turns out that this is true for Job Postings, and emphatically so - there exist hundreds of such datasets. Not all data sets are of equal value - many are very small and many have limited features - And as such I located and downloaded 5 adequately expressive datasets. The details of these datasets are expressed in Table A.1.

As can be seen, some datasets are created by individuals through web-scraping methods, while others are sample datasets from the much larger stores of commercial entities (e.g. Techmap[66]). Each of these data sets satisfy the minimal criteria for utility in this project. Ranging from a few hundred to a few thousand records, they all possess the required features:

- Date component: A timestamp of when the posting was created.
- Title component: The advertised job title.
- Description component: The main body of texts for a Job Posting.

Table 3.1: Overview of datasets used for Job Postings

Dataset Name	Date Range	Details	Source
1000 ML Jobs US	12/2023 - 04/2025	Mostly within March & April of 2025, all jobs are ML related	[60]
Linkedin Data Engineer Job Postings	17/12/2023 17/12/2023	- Extra "skills" column - This is appended to "description"	[61]
Data Science Job Postings & Skills (2024)	19/01/2024 21/01/2024	- Extra "skills" column - This is appended to "description"	[62]
LinkedIn Job Postings (2023 - 2024)	23/03/2024 20/04/2024	- -	[63]
Job Postings from Ireland (October 2022)	01/10/2022 31/10/2022	- Ireland only; Sample dataset from Techmap.io's commercial product	[64]
US Job Postings from 2023-05-05	05/05/2023 05/05/2023	- USA Only; Sample dataset from Techmap.io's commercial product	[65]

3.2.1.4 Cleaning & Collating

These five datasets cover different temporal (See Fig 3.2.1.4), regional and occupational (See Fig ??) ranges. This next stage involved reconfiguring their composition into a single representative dataset. Job Postings: Distribution of Timestamps

Job Postings: Distribution of Vacancy Titles As is the case for the majority of this project, this process was undertaken using **Python** (and the library **Pandas** - full information on utilized software & libraries is under Appendix A.1.3.2). Some datasets included an additional “Skills” field. During pre-processing, I concatenated this to the “Description” field so that all Job Postings are represented as a single textual body. This ensures consistent treatment across postings, whether or not they explicitly separate skills from the main description.

Thereafter, each dataset required varying degrees of processing to retrieve & represent the 3 necessary fields, which are thereafter compiled into one big **Pandas DataFrame**. Thereafter, I generated a new field - “Mapped Title” - which uses *regular expressions* to map the noisy “Title” field into a cleaner aggregation field. Using this, undesirable fields (i.e. not relating to AI or Data Science) were removed - this was done in alignment to the design choice as set out in Section ??, and the method is largely effective (assuming the proper **regex** definitions).

As a result, the full Job Postings dataset for downstream usage totals 18690 records ranging from 10/2022 → 04/2025.

3.2.2 OSS repositories

3.2.2.1 Overview of Data Landscape

The process for retrieving GitHub repositories data is much simpler in the availability sense - free API access is offered, via 2 alternative endpoints: REST API and GraphQL API. To reiterate, the features of this data channel required are the repository metadata (including the README, description, owner-name, repository-name, and topic-tags) and popularity data, representable by count of stargazers (a ‘stargazer’ of a repository is a user that has starred that repository) as a timeline of the respective creation-events (When the user ‘starred’ the repository). It is important to note that these are but a subset of the available data points that are collectible, and usable, from GitHub.

For example, it is possible that other forms of textual data exist on an ad-hoc basis, potentially providing more fine-grained technical information that is relevant to representing its ideas. Furthermore, other measures of popularity and of development evolution Would provide a more holistic understanding of a repository’s true development history. Nonetheless, the choices made are pragmatic in that they are, on the whole, sufficiently indicative of repository characteristics, for any repository.

3.2.2.2 Repository selection

A subset of repositories was required, as acquiring and processing all of GitHub is infeasible. Manual curation would be both laborious and subject to bias, so automated alternatives were sought.

I approach this in two different ways:

1. Web Scraping by GitHub topic
2. Informed by a Third-Party source

The first approach used seed topics (e.g. Machine Learning”, Natural Language Processing”), retrieving the top 30 repositories per topic and recursively expanding via their associated tags. This method effectively captures a broad subspace of related repositories, but it has drawbacks: niche or irrelevant topics may be included, and many relevant repositories lack topic tags altogether (this latter problem is exactly that as tackled with [52]).

The second approach therefore seeds from an auxiliary data source a list of repositories that are trending/relevant to sub-topics of software engineering / AI. This source, *OSS Insight* [67], is an analytics project that provides various functionalities, including daily-updated topical collections of repositories that are *trending*

or *popular*. Their algorithm for determining these repositories include ‘scores’ of stars/forks and a ‘base score’, computing optimized ratings from weighted treatment of such metrics. Their ranking algorithm is based on a composite score including stars, forks, and a base score, with weights optimized over these metrics. The exact details are not fully disclosed, though their open development process suggests that additional engagement signals may be incorporated (e.g. activity around pull requests and issues, and code evolution in line deletions/creations) [68]. Moreover, their GitHub repository reference that ‘collections’ are curated by a *community process*, wherein users can submit collections in pull requests. This constitutes a form of community-driven annotation, providing an effective low-time/effort alternative to developing a bespoke classification mechanism or manually labeling categories.

These repositories are externally validated as relevant to AI, reducing the need for subjective judgment on my part. All that is left to me is to select the names of “Collections” (analogous to the “seed topics” in my first approach) for usage, as seen in table ???. Further consideration is paid to the problems and limitations of this approach are detailed in section ???. For the purposes of manageability, this is an appropriate method of sampling a relevant set of repositories, and fits nicely into a programmatic pipeline by the simple API access provided.

Importantly, *OSS Insight* metadata (e.g. ‘Collection names’) is not used in downstream analyses, since it references data external to the repository contents. They are perhaps useful for categorization in an evaluative sense, but as they reference data extraneous to the repository contents itself, it would not be appropriate to treat this in kind.

3.2.2.3 Acquisition: Web Scraping

Initially, I chose to develop a simple web scraping method for pulling repository metadata. Though somewhat effective. This method presents major maintainability issues and furthermore is not necessarily the most compliant mechanism for acquiring website data.

3.2.2.4 Acquisition: GitHub’s REST API

Thereafter, I graduated to develop a script for acquiring the metadata of GitHub repositories via GitHub’s REST API endpoint. This method is not, however, extensible to observing the star-history for a project. Such data is most effectively stored (and accessed) using a Graph Database, relating as it does to the activity of users, affording the benefit of also storing other forms data inherent to the interconnected network nature of the GitHub (and broadly, OSS) ecosystem.

3.2.2.5 Acquisition: GitHub’s GraphQL API

Therefore, I developed a method to access GitHub’s GraphQL API endpoint, using defined graph-query-language queries that can effectively be managed/alterd in a separate part of the codebase. This method involves paginating through the set of sometimes hundreds of thousands of star-creation events for a repository, extracting the related time-stamps. At $\approx 1.5 \frac{\text{Seconds}}{\text{Request}}$ & retrieving 100 star-events per request, this process is time-intensive for large repositories (The largest in this project is **TensorFlow**’s repository, with $\approx 190,000$ stars).

To consolidate this process, the metadata acquisition script was refactored to also make use the GraphQL API. This results in a more robust, maintainable and adaptable component for the acquisition of GitHub repository data, which remains extensible to acquiring other data-points (e.g. Forks, Commits, Pull Requests, etc.).

3.2.2.6 Results

A full dataset of GitHub repositories and their related metadata & stargazer-history are therefore retrieved via maintainable and adaptable code. The script thereafter aggregates and stores stargazer history in different formats, namely wide and long formats:

- **Long:** A tabular file with fields of [`‘date’`, `‘repository’`, `‘stars’`].
- **Wide:** A tabular file resembling a matrix of shape (n, m) , where n is the number of repositories and m is the number of discrete time periods (daily, monthly, or quarterly). The time axis is defined as the range from the earliest star-event across all repositories ($\min D$) to the latest ($\max D$), partitioned into equal-length periods. Each entry (i, t) records the number of star-events for repository i in period t .

3.2.3 Auxiliary Resources

The term ‘Auxiliary’ here represents a channel of data that is both an assortment of sources that were used for evaluative (or otherwise informative) purposes and not central to the representation/analysis itself.

3.2.3.1 Keyword Dictionary

As is made clear in section 3.3, there is a need for some ‘ground truth’ measure against which representation methods can be evaluated.

In essence, there is a requirement for such abstract representation methods for *labelled* data that is not easily devised. This would entail definitively creating a metric of “similarity” between Job Posting/repository pairs $(R_r, J_j)^1$, a task that is difficult to define clearly, as well as being heavily time-intensive and laborious. To overcome this issue, a heuristic based method was developed that utilizes a basic *dictionary* of relevant terms/phrases for keyword matching. The curation of this data is integral to the methodology of the evaluation baseline (hereafter referred to as the **WeakModel**), so the details are left to section 3.3.

3.2.3.2 ESCO Skill Classification

The full ESCO Skills & Competences classification dataset was downloaded and used for exploring the contents of Job Postings.

This data is not utilized in any meaningful sense by the project’s pipeline - this is a methodological alignment with philosophy such as in [48], expressing a preference to rely on a data-driven & discovered taxonomy. While there is conceivable utility to the ESCO classification, it is not immediately clear that the sort of shared semantics between Job Postings and GitHub repositories would resemble their taxonomy, so their contents are left aside for this paper’s primary developments.

3.3 Weak Model Baseline

In supervised learning, evaluation is straightforward because we have a **target** or ground truth: we know the correct labels or outcomes, so we can measure how well our model’s predictions match reality. For similarity analysis, however, the notion of a target is less obvious. We need to define what it means for two items to be “similar” and establish a reference or standard against which our representation method can be judged.

Intuitively, this means constructing a set of examples (such as pairs of texts) where the degree of similarity is known or agreed upon—either through expert annotation, crowd-sourced judgments, or some other reliable method. These examples serve as the **benchmark** or target, allowing for systematic comparison of different representation methods to determine which one best captures the desired notion of similarity.

In this case, the problem lies with labels - there are none. It is therefore necessary to define labels before proceeding to evaluating the representation methods.

¹To aid notation and simpler discussion, an ambiguous pair of (Job Posting/repository) is (R_r, J_j) , where R_r and J_j refer to the r^{th} and j^{th} repository and Job Posting data-points respectively

The chosen method follows, broadly, fuzzy matching a dictionary of hand-crafted keywords.

3.3.1 Keyword Dictionary Design

- Construction of seed keyword list.
- Curation and refinement process.
- Matching rules.

3.3.2 Job–Repo Pair Labeling

- Description of how weak model generates positive/negative matches.

3.3.3 Purpose of Baseline

- Provides pseudo-ground truth for evaluation.

3.4 Representation Methods

3.4.1 Text Preprocessing

- Tokenization and filtering.
- Field selection (job titles vs repo descriptions).

3.4.2 Sentence Embedding Models

- Model choice (e.g., transformer-based embeddings).
- Input fields for embeddings.

3.4.3 Similarity Computation

- Cosine similarity between job and repository vectors.
- Production of job–repo similarity matrix.

Illustration of job–repository similarity matrix construction.
--

3.5 Time-Series Construction

3.5.1 repository Popularity

- Quarterly star count aggregation.
- Handling of older vs newer repositories (zero-padding).

3.5.2 Job–Repo Similarity Series

- Aggregation of similarity values across time.

3.5.3 Alignment of Series

- Aligning job–repo similarity time series with repository popularity series.
- Handling mismatched time horizons.

3.6 Evaluation Metrics

- Correlation coefficient: degree of linear relationship.
- Maximum cross-correlation and lag: lead–lag structure between series.
- Fraction of high-alignment periods: consistency of strong similarity.
- Popularity-adjusted similarity: weighting similarity by repo star counts.

Example illustration of correlation and lag metrics between repository popularity and job similarity series.

3.7 Summary

- Recap of pipeline.
- How each component contributes to testing the research question.
- Transition to Results/Analysis.

Chapter 4

Results

4.1 Evaluations of Experiments

4.2 Discussion

4.3 Problems & Limitations

Chapter 5

Critical Evaluation

In this chapter (it may not be Chapter 4 for you, but probably Chapter 6 or 7 once all the core chapters have been added.)

The critical evaluation consists of a discussion, leading to conclusion. It is an essential part of a master's degree.

It shows that you can not only carry out a substantial piece of work, but that you can reflect on it, and think critically about how you might have done it better.

Examiners view the critical evaluation as very important.

Critical evaluation should contain

- Strengths and weaknesses of your project
- If you were unable to attain any deliverables, then why
- What are the future plans for your project if you are to continue

You will be presenting this during demonstration but here you need to discuss them in details.

5.1 Assumptions

5.1.1 GitHub repository as proxies for OSS

5.1.2 Job Posting Source concatenation

Chapter 6

Conclusion

A brief summary of all that has gone before.

May include some directions for future work.

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Appendix A

A.1 Generative AI

This paper uses 3 Generative AI Tools:

1. **Cursor IDE** for expanding the weakly supervised model keyword dictionary, and implementing code efficiency improvements (using GPT-5) [69], [70]
2. **Elicit AI** to assist in literature exploration phase [71]
3. **ChatGPT** for advice, critique & education [70]

None of these tools were used to generate, or otherwise automate, the content of this paper (Except in the specific case of A.1.3.2) Underlying my use of generative AI was the imperative to avoid “Copy & Paste” strategies and instead use them as research, exploration and educational tools.

A.1.1 Elicit

For research, Elicit is a useful tool that performs a complex literature search based on detailed prompts. This was valuable for exploring research & informing the thesis more precisely.

I would, in future, make greater use of this tool for guiding the literature review stage.

Here’s are the *exact* prompts:

Prompt: Job Description Research

To what extent can we use the “Job description” section of Job Advertisements to map, forecast or understand the demand of certain skills or job requirements? A methodology that describes time-series representation of features in the Natural Language text, with NLP or other implementations.

Prompt: Innovation informed AI-Skill Forecast Research

Can innovation signal data (research papers, patents or open source software (e.g. GitHub)) improve the strength of skill demand forecasts derived from job postings (job boards like LinkedIn or Indeed), particularly for AI related skills? E.g. “prompt engineering” or “MLOps”

A.1.2 ChatGPT

For Code, ChatGPT was used for advice, debugging code or learning new programming frameworks. For writing, it was used as a critique assistant, to help explore ideas and critically assess the coherence of arguments or passages.

Here are some example prompts (adjusted, on a case-by-case basis):

Prompt: Conceptual Understanding

*Advise me on how to think about **{Concept/Tool/Framework/Method}** conceptually. I'd like to know the mental models or guiding intuitions that professionals use.*

Prompt: Conceptual Understanding

*Below is my summary of what I currently understand about **{Concept/-Tool/Framework/Method}**. Please critique it: point out what is accurate, what is incomplete, and where I've misunderstood. The goal is to refine my own understanding, not to receive a polished replacement text.*

Much of the time I use ChatGPT for workflow and tool advice, especially with regards to using LaTeX:

Prompt: Workflow advice

*I'm setting up a research workflow using **{Tool/Method}**. Can you suggest general strategies, potential pitfalls, and best practices? My goal is to design the workflow myself, but I'd like conceptual guidance.*

Prompt: L^AT_EXHelp

Advise me on strategies for managing a large dissertation in LaTeX. For example, what are best practices for splitting chapters into separate files, managing bibliographies, and handling figures/tables so that the project remains maintainable?

Prompt: Code Critique

*Below I include a piece of code. Rather than rewriting it for me, can you summarize your understanding of its purpose, critique its efficiency, point out any readability/-maintainability issues, and suggest what **principles** I should look at to improve it?*
{Code}

Prompt: Writing Critique

*I am writing a dissertation in **{Dissertation Context}**. I will paste a passage of my draft below. Please provide a detailed critique focusing on:*

- 1. clarity of expression and readability,*
- 2. academic style and tone,*
- 3. logical flow and argument structure, and*
- 4. any redundancy, ambiguity, or opportunities for conciseness.*

*Give only high-level feedback (structure, argument, coherence) - do not provide any specific rephrasing or line edits. Assume I am aiming for a professional, dissertation-quality standard. Here is the passage: **{Text Passage (No more than a paragraph at a time)}***

Each of the above prompts are used as guiding structures, and much of the time were adapted in scenario-specific follow-ups.

A.1.3 Cursor

A.1.3.1 Code Improvements

Cursor was used to advise on code improvements that were made strictly *after* a working solution was in place. The purpose of this was to improve both the computational efficiency and readability of the codebase. Further, it serves as an educational tool for improving my coding skills.

These adjustments were not made using the “Agent” feature of Cursor, but with “Ask”, such that it provides its suggestion for implementation at my discretion.

Here’s an example prompt for this use-case:

Prompt: Code Improvement Suggestions

*Within **{Context File}** in lines **{Context Lines}** is a piece of code. Currently it is functional. Rather than rewrite it for me, can you summarize your understanding of its purpose, critique its efficiency, point out any readability/maintainability issues, and suggest what **principles** I should look at to improve it?*

A.1.3.2 Weak-Model Keyword Generation

This is sole exception case where generated content was used: Instructing Cursor (using GPT-5) to construct the keyword dictionary for the weakly-supervised model. Its task here was to (In iterations of 50 items; 4 runs → 200 items):

- Expand a *Seed List* of keywords (Obtained by iteratively scraping GitHub topics)
- Match keywords to their commonly used acronyms
- Categorize the keywords

The grounding of this implementation in a *Seed List* was **instrumental** in making this effort defensible.

The purpose of this was to reduce manual & laborious data entry work - it should be noted that the only practical advantage of this implementation was to save time and effort. This entire process was audited and refined manually thereafter, to ensure output quality.

Prompt: Keyword List Curation

In {Context File} you'll find a list of keywords relating to Artificial Intelligence & related GitHub repository topics. Your task is to expand this keyword list, to a total of no more than 50. The resulting list should contain core concepts, tools, programming languages and domain-specific terms. If a keyword is commonly referred to by an acronym, these should be paired with its relevant verbose version. The acronym field should be left null for terms without commonly used acronyms. Be sure to preserve the casing for each keyword - e.g. "pytorch" should be "PyTorch". Make sure not to duplicate existing entries.

Prompt: Categorization

Within {Context File} are keywords that relate to artificial intelligence (including their acronyms). Create a new field of categories for each keyword into a set of no more than 25 different categories. These categories should be informative, and concise; if deemed necessary, one term can belong to multiple categories (no more than 3).

Data Sources

Third Party Code and Software Libraries

1. Whatever Software I end up using

A.1.4 ESCO Skill Classification

This publication uses the ESCO classification of the European Commission.

Table A.1: Overview of datasets used for Job Postings

Dataset Name	Date Range	Details	Source
1000 ML Jobs US	12/2023 - 04/2025	Mostly within March & April of 2025, all jobs are ML related	[60]
Linkedin Data Engineer Job Postings	17/12/2023 17/12/2023	- Extra "skills" column - This is appended to "description"	[61]
Data Science Job Postings & Skills (2024)	19/01/2024 21/01/2024	- Extra "skills" column - This is appended to "description"	[62]
LinkedIn Job Postings (2023 - 2024)	23/03/2024 20/04/2024	- —	[63]
Job Postings from Ireland (October 2022)	01/10/2022 31/10/2022	- Ireland only; Sample dataset from Techmap.io's commercial product	[64]
US Job Postings from 2023-05-05	05/05/2023 05/05/2023	- USA Only; Sample dataset from Techmap.io's commercial product	[65]

6.2 Word Counts

Category	Count
Words in text	11261
Words in headers	234
Words in captions	57
Tables/Figures	6
Inline math	46
Displayed math	0

Table 6.2: Document summary

File	Text	Headers	Captions	Floats	Inline Math	Displayed Math
report	0	16	0	0	0	0
./envs/argument-environment	121	0	0	0	0	0
abstract	39	0	0	0	0	0
preface	171	0	0	0	0	0
introduction	666	7	0	0	0	0
literature	6607	92	0	0	21	0
methodology	2619	77	0	0	22	0
results	0	7	0	0	0	0
critical-evaluation	134	11	0	0	0	0
conclusion	16	1	0	0	0	0
appendices	1002	21	0	0	3	0
./figures/feedback-loop	0	0	0	1	0	0
./figures/pipeline	0	0	0	1	0	0
./figures/job-data-time-variation	3	0	0	0	0	0
./figures/job-data-title-variation	4	0	0	0	0	0

Table 6.3: Word count by file (`texcount`)